

# Guduru Praneeth

## AI/ML ENGINEER

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### SUMMARY

B.Tech Computer Science (AI & ML) graduate experienced in building end-to-end AI systems across RAG pipelines, LLM fine-tuning, agentic AI, and classical machine learning. Skilled in Python, LangGraph, FastAPI, PyTorch, and scikit-learn, with hands-on experience developing and evaluating AI solutions for real-world applications.

### SKILLS

**Languages & Core:** Python, SQL, Java(Basics)

**Generative AI:** RAG, LLMs, LangChain, LangGraph, LLM Fine-tuning (LoRA/QLoRA), Prompt Engineering

**Machine Learning:** PyTorch, Scikit-learn, Computer Vision, PCA, SVM

**Tools & Deployment:** FastAPI, Streamlit, ChromaDB, Hugging Face, AWS, Git, Docker(Basics)

### EXPERIENCE

**AI/ML Intern — Retearn Technologies Private Limited Hyderabad |**

Aug 2025 – Oct 2025

- Built image classification model using PCA and SVM to identify waste types from near-infrared (NIR) images
- Achieved 89.2% accuracy by optimizing model parameters on 2,100+ images per class
- Used cross-validation and evaluation metrics (precision, recall, F1-score) to ensure model reliability
- Annotated around 10000 images for data preparation.

### PROJECTS

#### 1. DataPilot – Multi-Agent Natural Language SQL System

Python • LangGraph • FastAPI • Streamlit • MCP

- Built a multi-agent system that converts natural-language questions into executable, schema-aware SQL queries.
- Designed an agent workflow covering query parsing → SQL generation → validation → execution → insight generation.
- Built a full-stack interface enabling non-technical users to query relational databases using natural language.

#### 2. Veritas – Self-Correcting RAG System

Python • LangGraph • FastAPI • ChromaDB • Streamlit

- Engineered a self-correcting RAG pipeline over 150+ ML research papers, combining retrieval, answer generation, and validation.
- Optimised retrieval latency from 2.8s to 450ms using HNSW indexing and embedding caching.
- Evaluated retrieval and generated responses using relevance and correctness benchmarks, achieving 5/5 average evaluation scores.

#### 3. Finance LLM Fine-Tuning with QLoRA

Python • PyTorch • Hugging Face • PEFT

- Fine-tuned Qwen2-1.5B on finance Q&A data using QLoRA and 4-bit quantization on an RTX 3050 4GB GPU.
- Reduced evaluation perplexity from 21.2 to 6.6 through parameter-efficient fine-tuning.
- Trained 4.6% of model parameters using LoRA adapters while keeping the base model quantized.

#### 4. AutoMLEngineer

Python • Scikit-learn • XGBoost • Pandas

- Built an automated ML pipeline covering data cleaning, feature engineering, model selection, hyperparameter tuning, and report generation.
- Evaluated 8 classification and 10 regression algorithms using RandomizedSearchCV.
- Automated model evaluation reports with performance metrics and feature-importance analysis.

### EDUCATION

**Malla Reddy University, Hyderabad**

B.Tech Computer Science & Engineering (AI & ML)

Graduated: June 2026 | **CGPA: 8.84/10**

### CERTIFICATIONS AND ACHIEVEMENTS

- RAG for Enhanced AI Outputs — IBM
- Model Context Protocol (MCP) — Anthropic
- AWS Cloud Practitioner Essentials — AWS
- Introduction To Machine Learning — CourseEra
- AI Fluency: Framework & Foundations — Anthropic
- 1st Place, Intelithon Hackathon (~300 participants) for building HeartHealthAI a pre